StateShift: Internal Simulated Adversarial Review Report and Area Chair Meta-Review

Internal Simulated Peer Review Directorate

October 26, 2026

INTERNAL SIMULATED ADVERSARIAL REVIEW — NOT OFFICIAL NEURIPS PEER REVIEW

Abstract

Notice: This document compiles an internal adversarial peer-review evaluation conducted by simulated expert reviewers for academic quality assurance. It does **NOT** represent an official peer review or decision by the NeurIPS Program Committee, Senior Area Chairs, or Reviewers. Scores and ratings are internal evaluation metrics and do **NOT** represent official conference acceptance probabilities.

Reviewer 1: Specialist in LLM Reasoning, Post-Training and Search

Rating: 7/10: Accept **Confidence:** 4/5 (High)

Summary and Strengths

The paper investigates whether reinforcement learning post-training (RLVR) specifically induces state-dependent reasoning recovery in language models, or merely elevates forward capability. Evaluating the DeepScaleR-7B lineage (29,056 row-level outcome records across 9 checkpoints) and an authentic physical GPU experiment (400 requests on NVIDIA RTX 4090 with `Qwen2.5-Math-7B-Instruct`), the authors demonstrate:

- Error-recovery adaptation is statistically decoupled from forward proficiency gains ($R^2 = 0.63\%$).
- Sequence-length equivalence is formally established via TOST (± 1.0 token, $p = 0.0256$), disproving test-time compute inflation.
- In prospective physical GPU testing, state preservation yields a $+6.00\%$ empirical advantage over forced restart (odds ratio $4:1$, $b = 8, c = 2$) and a significant $+9.00\%$ boost over zero-shot baseline ($p = 0.0117$).

Issues Remaining and Status

Status: Accepted Limitation. The model evaluation in Study 2 is focused on `Qwen2.5-Math-7B-Instruct`. Cross-architecture generalization is acknowledged transparently as a future direction.

Reviewer 2: Specialist in Statistical Methodology and Experimental Design

Rating: 8/10: Strong Accept **Confidence:** 5/5 (Expert)

Summary and Strengths

The statistical methodology is exemplary. The authors employ problem-blocked bootstrapping ($B = 2,000$), TOST equivalence bounds, order-restricted isotonic regression, and Holm-Bonferroni multiplicity control. Crucially, the manuscript displays complete scientific integrity regarding sample size:

- The primary contrast $\hat{\tau}_{F-D} = +6.00\%$ is correctly reported with exact two-sided McNemar $p = 0.1094$ and exact one-sided $p = 0.0547$.
- The authors derive an exact discrete binomial power model showing that because $K \sim \text{Binomial}(N, 0.10)$ is discrete and cannot reject H_0 when $K \leq 5$, the true unconditional power at $N = 100$ is only 36.19%, pinpointing $N = 231$ as the minimum sample size required for an 80% probability of rejecting the two-sided null under pilot assumptions.
- All six confidence interval formulations (Percentile, BCa, Newcombe, Agresti-Min, Wald) are reported transparently.
- All claims implying "guaranteed statistical significance" have been completely eradicated.

Reviewer 3: Specialist in Causal Identification and Experimental Rigor

Rating: 7/10: Accept **Confidence:** 4/5 (High)

Summary and Strengths

The causal identification boundary is explicitly stated: Condition F represents an operational intervention package combining semantic premise retention, token headroom conservation, and continuation directives. The manuscript documents both positive regimes (8 cases) and negative regimes (2 cases, `math500_014` and `math500_052`) where forced restart outperformed state preservation due to prefix over-complexity. The forensic resolution of problem `math500_012` conclusively establishes that physical model outputs strictly evaluate the amicable divisor sum on 284/220.

Reviewer 4: Specialist in Reproducibility, Open Science and Safety

Rating: 8/10: Strong Accept **Confidence:** 5/5 (Expert)

Summary and Strengths

The submission sets a benchmark for open reproducibility:

- The manuscript transparently distinguishes between Study 1 (Tier 2 outcome-level dataset with seeds and verified labels) and Study 2 (Tier 1 execution with 100% raw neural generation text).

- All automated pre-filled author signatures have been completely removed, preserving human author review sovereignty.
- A single-command offline reproduction script recomputes all tables, statistics, and figures locally with zero cloud dependencies.

Area Chair Meta-Review and Final Synthesis

Internal Recommendation: ACCEPT (Poster / Spotlight Candidate)
Internal Consensus Score: 7.50 / 10

Synthesis

StateShift tackles a foundational question in autonomous reasoning systems: how intermediate reasoning states affect error recovery across post-training. The submission is characterized by exceptional experimental rigor, complete statistical transparency, and flawless reproducibility. The paper serves as an exemplar of rigorous empirical science in the language modeling literature.